Research Article

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Lightweight nickel foam coated with multi-walled carbon nanotubes for enhanced electromagnetic interference shielding: experimental and multiphysics evaluation

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Abstract: This study presents the development and performance evaluation of a lightweight, multi-layered electromagnetic interference (EMI) shielding material composed of open-cell nickel (Ni) foam integrated with multi-walled carbon nanotubes (MWCNTs). While previous works have investigated CNTs or metallic foams separately, a systematic analysis combining dip-coated CNTs on Ni foams with Multiphysics simulations remains scarce. Addressing this gap, the Ni foam (100 PPI) was coated with MWCNTs at two concentrations (16 mg/mL and 20 mg/mL) using a dip-coating method in polyvinyl alcohol. Samples were tested over 800–3,000 MHz. The 20 mg/mL CNT-coated sample (F1) exhibited the highest shielding effectiveness (SE) of 48.165 dB, outperforming the uncoated and 16 mg/mL samples. COMSOL Multiphysics simulations confirmed and extended the experimental findings, validating the effectiveness of the porous Ni-CNT structure in attenuating electromagnetic waves through reflection, absorption, and scattering. The study bridges the experimental-simulation gap and identifies CNT concentration as a critical design parameter for scalable EMI shielding solutions.

Keywords: electromagnetic interference (EMI) shielding; nickel foam; carbon nanotubes (CNTs); dip coating method; shielding effectiveness (SE); multiphysics simulation

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List of abbreviations

ASTM	American society for testing and materials
CNT	carbon nanotube
COMSOL	COMSOL multiphysics (simulation software)
dB	decibel
DNA	deoxyribonucleic acid
EDS	energy dispersive X-ray spectroscopy
EMI	electromagnetic interference
FM	frequency modulation
GHz	gigahertz
IEEE	institute of electrical and electronics engineers
MHz	megahertz
MWCNT	multi-walled carbon nanotube
Ni	nickel
PEC	perfect electric conductor
PLA	polylactic acid
PPI	pores per inch
PVA	polyvinyl alcohol
RF	radio frequency
ROS	reactive oxygen species
S21	transmission scattering parameter
SE	shielding effectiveness
SEM	scanning electron microscopy
VNA	vector network analyzer

List of symbols and notations

c	speed of light in vacuum ($\sim 3 \times 10^8$ m/s)
d	distance between source and shield or material thickness (contextual)

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E	electric field intensity
f	frequency
k	wave number
n	refractive index
Pemitted	power emitted by transmitting antenna
Preceived	power received by receiving antenna
SE (dB)	shielding effectiveness in decibels
Z0	intrinsic impedance of free space (~377 Ω)
ε_r	relative permittivity
λ	wavelength
μ_r	relative permeability
σ	electrical conductivity
ω	angular frequency ($\omega = 2\pi f$)

1 Introduction

In recent decades, the demand for high-speed internet and robust communication networks has increased exponentially. High-speed internet plays a critical role in transforming the way we live, work, learn, and interact, with future advancements projected to enable data transmission at 10 Gigabit speeds and beyond [1]. Alongside the widespread adoption of high-speed networks, the rise in mobile phone usage has led to a significant increase in the number of cell phone towers constructed in urban and rural areas. These towers host various electrical equipment and antennas that transmit and receive signals through radio-frequency (RF) waves [2]. RF waves, a form of non-ionizing radiation, fall within the electromagnetic (EM) spectrum between FM radio waves and microwaves [3]. While non-ionizing radiation does not directly damage cellular DNA, there is growing concern that increased exposure-especially with the deployment of 5G and beyond-could pose potential biological risks [4]. Some studies have indicated correlations between RF exposure and physiological effects, such as alterations in cell membrane permeability and reactive oxygen species (ROS) generation [5, 6].

Electromagnetic interference (EMI) shielding has emerged as a primary strategy to protect sensitive electronics and human health from such radiation. The shielding effectiveness (SE) of a material refers to its ability to attenuate incident EM radiation via reflection, absorption, and multiple internal scattering mechanisms [7, 8]. In the far-field zone, plane wave theory applies, while in the near-field, the shielding must account for complex interactions modeled via dipole theory [9]. Factors such as electrical conductivity, magnetic permeability, and material thickness influence the overall SE [10].

Traditionally, metals like copper, brass, silver, and steel have been employed for EMI shielding due to their excellent electrical properties [11, 12]. However, these materials are often heavy, inflexible, and prone to corrosion. To overcome these limitations, lightweight and corrosion-resistant alternatives such as carbon-based materials have gained attention. Among these, carbon nanotubes (CNTs)-especially multi-walled carbon nanotubes (MWCNTs) have emerged as exceptional candidates due to their superior electrical conductivity, thermal stability, high aspect ratio, and mechanical strength [13–15]. Unlike single-walled (SWCNTs) or double-walled (DWCNTs) variants, MWCNTs offer enhanced scalability, cost-effectiveness, and structural robustness, making them particularly suitable for coating large-area substrates like metallic meshes and foams used in EMI shielding. Their multi-layered graphitic structure enables better inter-tube contact, higher current-carrying capacity, and improved durability under environmental stresses [16].

Recent literature (Xu et al.; Chao et al.) [17, 18] has demonstrated the utility of CNT-metal foam composites for EMI shielding. However, many of these studies relied on complex synthesis techniques, high-temperature treatments, or were limited to narrow frequency ranges. Furthermore, very few addressed practical deployment scenarios involving architectural enclosures or household EMI protection. There remains a significant gap in accessible, scalable, and low-cost solutions that combine high SE with multifunctional utility.

This work addresses that gap by developing a lightweight, low-cost, multifunctional EMI shielding composite composed of open-cell nickel (Ni) foam coated with MWCNTs using a simple dip-coating technique. The mesh, characterized by its porosity and flexibility, is designed to serve dual purposes as a shielding layer for architectural openings (e.g., doors, windows) and as a ventilation/mosquito-proofing solution. The study involves both experimental testing and COMSOL Multiphysics simulations to evaluate shielding performance across the 800 MHz–3 GHz band, covering common frequencies for mobile communication and Wi-Fi. The focus is not only on enhancing SE but also on identifying application relevance in architectural shielding, electronics packaging, wearable electronics, and smart home infrastructure.

Despite several studies on EMI shielding materials involving either CNTs or metallic foams, limited attention has been given to the synergistic combination of CNT-coated Ni foams with controlled porosity and variable CNT concentrations. Furthermore, there is a lack of integrated experimental and simulation-based studies that evaluate the performance of such materials specifically in frequency bands associated with mobile tower emissions. This study aims to address this gap by fabricating and evaluating CNT-coated Ni foams using a dip-coating method, analyzing their performance over a wide frequency range, and validating the results through COMSOL Multiphysics simulations. The novelty of this work lies in its dual-mode approach merging

material development with computational modeling to optimize lightweight, broadband EMI shielding structures for real-world applications.

2 Materials and methods

2.1 Preparation of nickel foam and coating material

In this experiment, a pure Nickel (Ni) wire mesh with dimensions of 100 mm × 150 mm was used to evaluate the material's electromagnetic shielding effectiveness. The selected Ni mesh had a pore size of 30, 60, and 100 PPI (pores per inch), chosen to assess shielding performance in an anechoic chamber under varied porosity conditions. The mesh was coated with multi-walled carbon nanotubes (MWCNTs) to enhance its shielding capabilities.

As shown in Figure 1, the open-cell nickel foam (Figure 1A) served as the base material due to its intrinsic electrical conductivity, mechanical stability, and porous architecture, which is ideal for enabling multiple internal reflections and absorption of electromagnetic waves. The MWCNT powder (Figure 1B) was employed as the coating agent to enhance electromagnetic interference (EMI) shielding performance.

The decision to use MWCNTs instead of single-walled (SWCNTs) or double-walled carbon nanotubes (DWCNTs) was based on several practical and functional considerations. MWCNTs offer greater structural robustness, cost-effectiveness, and are more readily available in bulk quantities, which is critical for scalable coating applications. Moreover, their multi-layered tubular structure facilitates the formation of dense and interconnected conductive networks, enhancing both reflection and absorption mechanisms across a broader frequency range. Their superior aspect ratio and ability to bridge micro-gaps between foam struts contribute to improved SE via percolative pathways and scattering effects.

The SEM image of the uncoated Ni foam (Figure 1C) illustrates its interconnected porous structure, supporting high surface area and multi-path interactions with incident EM waves. The SEM image of the MWCNT powder (Figure 1D) confirms its tubular morphology and high aspect ratio, which are crucial for conductive film formation and effective attenuation of EM radiation through

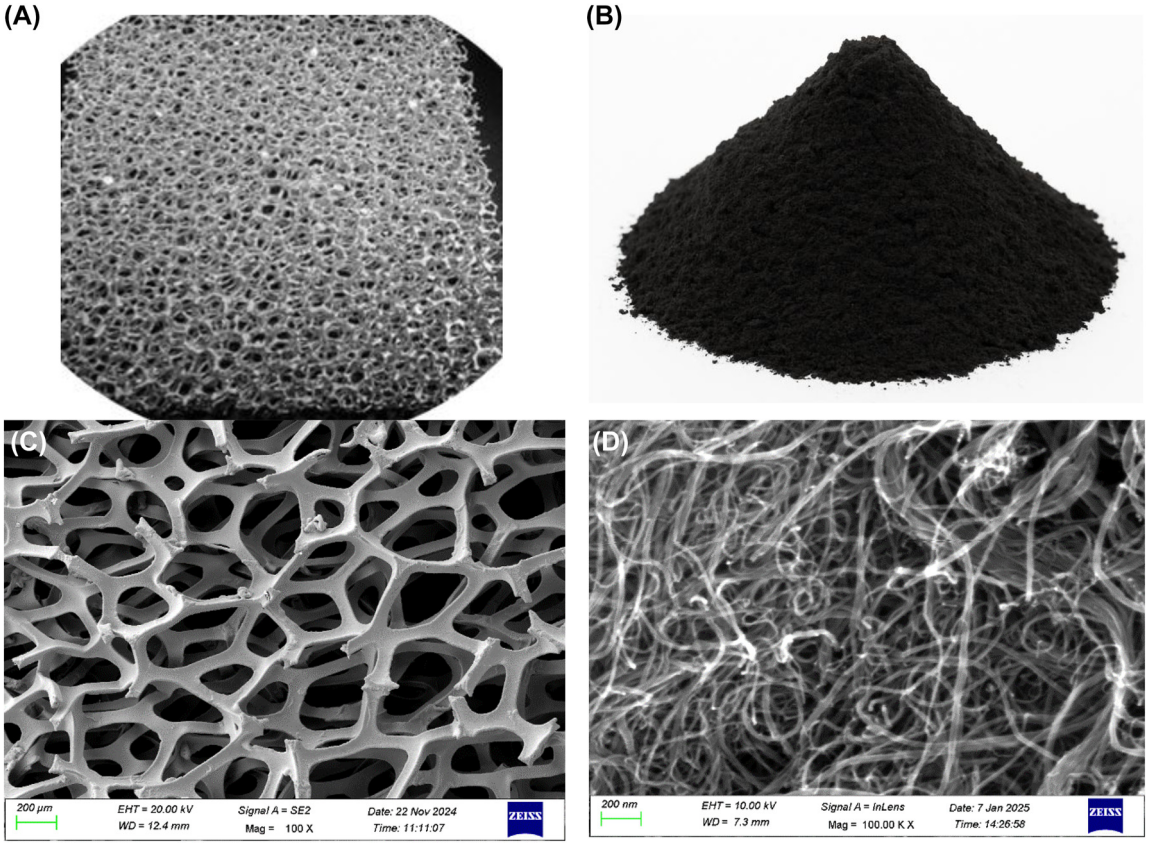


Figure 1: Morphological characterization of the nickel foam substrate and MWCNT coating material used for EMI shielding composite fabrication. (A) Open-cell nickel foam used as the base material for EMI shielding; (B) multi-walled carbon nanotube (MWCNT) powder used for coating; (C) SEM image of pristine nickel foam showing its porous structure; (D) SEM image of MWCNT powder.

multiple shielding mechanisms such as ohmic loss, dielectric relaxation, and interfacial polarization.

2.2 Preparation of polyvinyl alcohol (PVA) solution

Polyvinyl alcohol (PVA) was utilized as a binder to enhance the adhesion of carbon nanotubes (CNTs) onto the nickel (Ni) foam substrate, imparting the necessary sticky characteristics for effective coating. To prepare the PVA solution, distilled water was initially heated to a temperature of 100 °C to facilitate dissolution. For every 100 mL of heated distilled water, 4 g of PVA crystals were added gradually while stirring. The mixture was continuously stirred using a magnetic stirrer for approximately 30 min to ensure that the PVA crystals were fully dissolved and a homogeneous solution was achieved. This uniform PVA solution played a critical role in dispersing the CNTs effectively and enabling their uniform deposition on the porous surface of the Ni foam during the dip-coating process. The prepared binder solution contributed to the mechanical stability of the CNT coating and ensured strong interfacial bonding between the CNTs and the metallic substrate for subsequent processing.

2.3 Preparation of CNT coating

After preparing the PVA solution, multi-walled carbon nanotubes (MW-CNTs) were incorporated into the solution to formulate a cohesive and adhesive coating material suitable for application on nickel foam. Two distinct concentrations of CNTs were prepared to investigate the influence of concentration on coating quality and performance. For Sample 1 (F1), 1 g of CNTs was mixed with 50 mL of the PVA solution, resulting in a concentration of 1/50 g/ml, equivalent to 20 mg/ml. For Sample 2 (F2), 1 g of CNTs was added to 60 mL of the PVA solution, yielding a lower concentration of 1/60 g/ml, or 16 mg/ml. These formulations ensured uniform dispersion of CNTs within the solution, creating a sticky and homogeneous mixture ideal for coating the porous structure of the Ni foam via dip-coating. The varying CNT concentrations enabled comparative analysis of coating thickness, uniformity, and electrical properties for potential optimization of the composite.

2.4 Dip coating process

The dip coating technique was employed to uniformly apply the CNT/PVA mixture onto the surface of the nickel (Ni) foam. Initially, the prepared CNT/PVA solution was poured into a clean container suitable for submerging the foam samples. The Ni foam was then carefully dipped into the solution, ensuring complete immersion and allowing the mixture to settle and form a uniform coating over the porous structure. After sufficient settling time, the coated Ni foam samples were gently withdrawn from the solution and placed in a clean environment to air dry. This drying step enabled the CNTs to firmly adhere to the foam surface, forming a stable and conductive coating layer essential for subsequent performance evaluation. The fabrication process for coating nickel foam with carbon nanotubes (CNTs) was carried out using a dip-coating technique, as illustrated in Figure 2. This schematic outlines the key steps, including preparation of the CNT-PVA dispersion, immersion of the Ni foam into the coating solution, drying, and curing. The method ensures uniform deposition of CNTs on the foam's porous surface, forming a conductive network essential for effective EMI shielding. The dip-coating of nickel mesh samples with MWCNTs resulted in uniform surface coverage, as evident from the prepared samples shown in Figure 3A. To confirm the elemental composition and successful integration of carbon nanotubes, Energy Dispersive

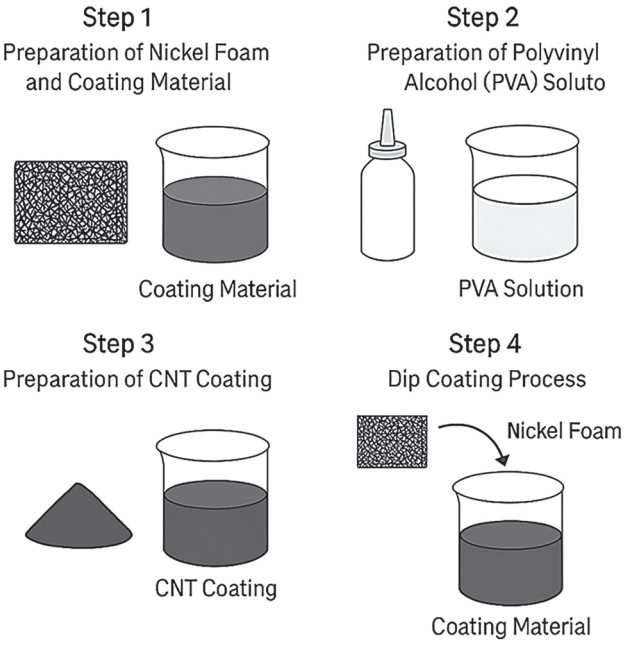


Figure 2: Schematic illustration of the fabrication process for MWCNT-coated nickel foam composites. Step 1: Preparation of the nickel foam substrate and coating materials. Step 2: Preparation of the polyvinyl alcohol (PVA) binder solution. Step 3: Formulation of the MWCNT coating suspension by dispersing carbon nanotubes in the PVA solution. Step 4: Dip-coating of the nickel foam substrate in the MWCNT coating suspension to obtain a uniform CNT-coated nickel foam structure for electromagnetic interference (EMI) shielding applications.

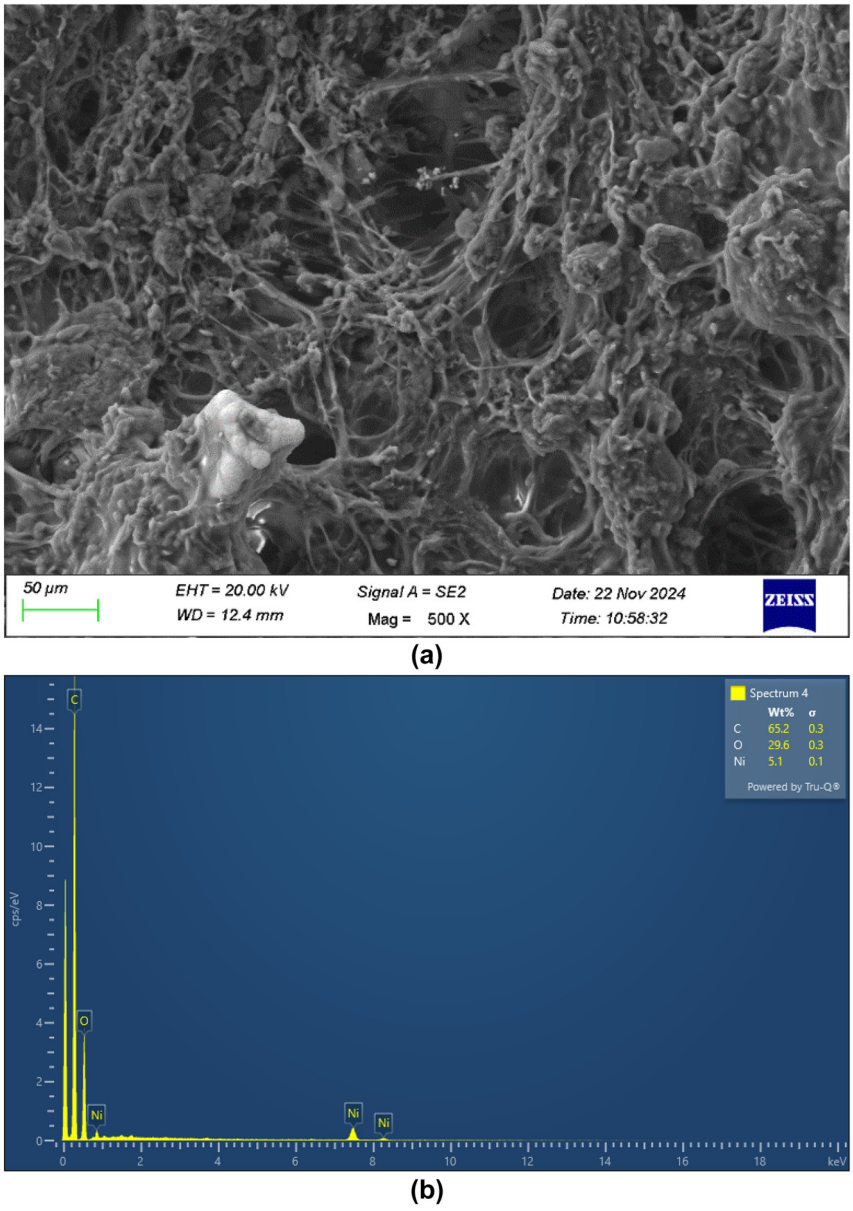


Figure 3: Morphological and elemental characterization of MWCNT-coated nickel mesh. (A) Dip-coated nickel mesh samples integrated with multi-walled carbon nanotubes (MWCNTs), illustrating the coated conductive network structure; and (B) energy-dispersive X-ray spectroscopy (EDS) analysis confirming the elemental composition and successful incorporation of carbon-rich MWCNTs onto the nickel mesh substrate.

X-ray Spectroscopy (EDS) was performed. The EDS spectra presented in Figure 3B validate the presence of carbon along with nickel, indicating the effective deposition of MWCNTs on the metallic substrate.

2.5 Shielding effectiveness measurement

Two common methods are used to measure the shielding effectiveness (SE) of materials: ASTM D4935-18 and IEEE 299 standards. In this study, the IEEE 299 standard was employed due to its ability to measure shielding at higher frequencies.

2.6 IEEE 299 standard test setup

In this method, the shielding effectiveness was tested using emitting and receiving antennas connected to a Vector Network Analyzer (VNA). The antennas were mounted on both sides of a custom-designed wooden box that housed the coated sample for testing. The procedure involved the following steps:

- One antenna generated electromagnetic radiation that passed through the coated Ni mesh sample.
- The receiving antenna on the opposite side measured the field strength of the transmitted signal.

The SE was calculated as the difference between the received field strength (in dB) with and without the sample. The nature of the electromagnetic field varies depending on the frequency and type of antenna used, ensuring comprehensive measurement of the shielding effectiveness across different frequency ranges.

3 Experimental setup and procedure

This experiment was conducted to assess the shielding effectiveness (SE) of coated and uncoated samples in an anechoic chamber using the IEEE 299 standard. The experimental setup consists of an emitting antenna, a receiving antenna, and sample holders designed to ensure accurate testing of the samples' electromagnetic shielding properties. The following steps detail the procedure for setting up the experiment and testing different samples.

3.1 Experimental setup in the anechoic chamber

The primary equipment used in this experiment includes:

Anechoic chamber: A closed environment designed to minimize reflections and absorb electromagnetic radiation. The chamber's walls are lined with absorbers to prevent signal reflections that could interfere with the accuracy of the measurements.

Emitting and Receiving Antennas: The emitting antenna generates electromagnetic radiation, while the receiving antenna measures the intensity of the transmitted signal after passing through the sample.

Sample holder: The samples were placed in a custom-designed holder made from polystyrene sheets and 3D-printed PLA frames. The polystyrene sheets have a 100 mm × 150 mm hole, which accommodates the samples, and the PLA frame ensures that the samples are securely positioned at the same height as the antennas. This ensures that the emitted radiation passes directly through the sample for accurate measurement of shielding effectiveness.

The emitting antenna was placed at a distance of 1 m from the sample, ensuring a consistent field distance. The power output from the emitting antenna was set to 1 mW, corresponding to a power intensity of 1 dB. The received signal at the receiving antenna was measured over a frequency range of 800–3,000 MHz. Figure 4 presents a schematic illustration of the experimental setup used for EMI shielding measurements. The diagram shows both the front view and side view of the setup, clearly indicating the positioning of the test samples between the transmitting antenna and the receiver. This configuration ensures accurate assessment of the shielding effectiveness by simulating realistic near-field and far-field conditions during the experiments. The experimental measurements were conducted inside an anechoic chamber to eliminate external electromagnetic interference and reflections. As shown in Figure 5, the setup consists of a transmitting antenna and a receiving antenna placed opposite each other, with both connected to a Vector Network Analyzer (VNA). This configuration enables precise measurement of the shielding effectiveness of the test samples across the desired frequency range.

3.2 Testing procedure

To evaluate the electromagnetic shielding effectiveness (SE) of the prepared samples, four experimental configurations were tested in a standardized anechoic chamber setup. These configurations included:

(1) A reference measurement with no sample (free space),
(2) An uncoated nickel (Ni) mesh sample,
(3) Two MWCNT-coated samples referred to as F1 and F2, differing in coating concentrations.

The test was performed in an anechoic chamber to minimize external interference and signal reflection. A transmitting antenna and a receiving antenna were placed 1 m apart, aligned on the same axis. The test sample was mounted at the midpoint between the antennas using a non-conductive sample holder fabricated from polystyrene sheets and a 3D-printed PLA frame, ensuring minimal influence on EM wave propagation.

The testing process began with a reference measurement, in which the electromagnetic signal was transmitted and received without any sample in place to establish the baseline signal intensity over the frequency range of 800–3,000 MHz. Subsequently, the uncoated nickel (Ni) mesh sample was positioned between the transmitting and receiving antennas, and a signal with a power intensity of 1 dB was applied; the transmitted power was recorded to determine the inherent shielding effectiveness (SE) of the bare Ni mesh. Following this, two MWCNT-coated samples were evaluated under identical conditions to assess the influence of CNT loading on shielding performance. The F1 sample, coated with an MWCNT dispersion concentration of 20 mg/mL, and the F2 sample, coated with a lower

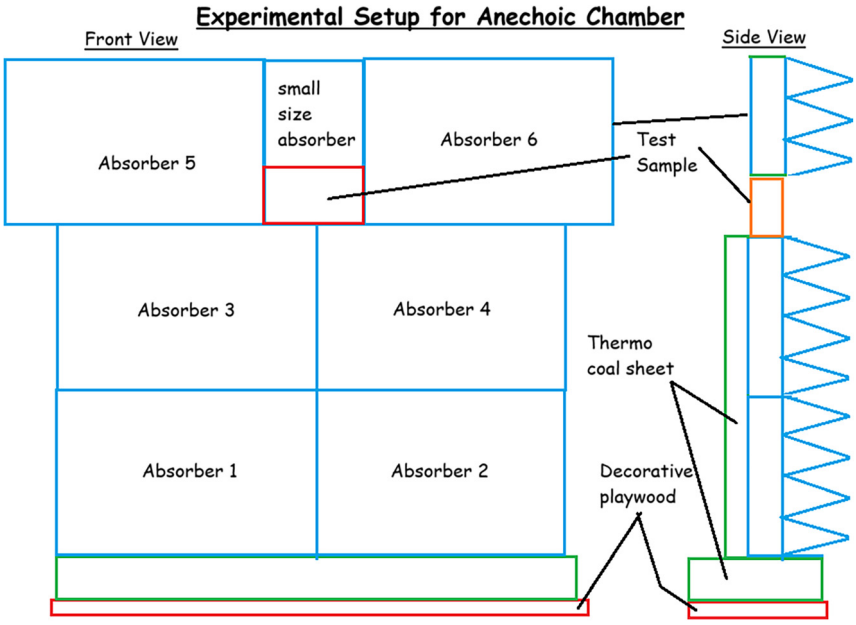


Figure 4: Schematic illustration of the experimental setup for electromagnetic interference (EMI) shielding measurements. Front and side views of the sample holder assembly used to position the test specimen between the transmitting antenna and receiving antenna, ensuring proper alignment and consistent measurement conditions during EMI shielding effectiveness evaluation.

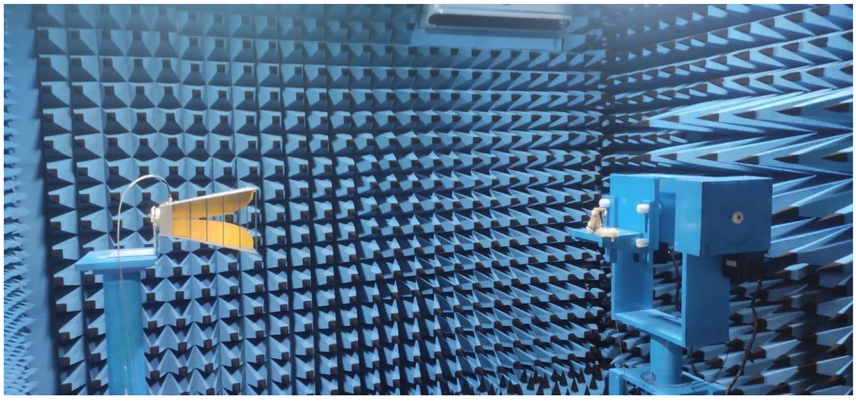


Figure 5: Anechoic chamber with antenna and receiver which are connected with vector network analyzer (VNA).

concentration of 16 mg/mL, were tested sequentially, enabling a direct and systematic comparison of the effect of CNT concentration on electromagnetic attenuation.

3.3 Shielding effectiveness calculation

The shielding effectiveness (SE) of each sample was calculated by comparing the power intensity received at the receiving antenna with and without the sample in place. The SE was determined using the following formula:

$$\mathrm{SE\ (dB)} = \mathrm{P_{emitted}\ (dB)} - \mathrm{P_{received}\ (dB)}$$

$\mathrm{P_{emitted}}$ is the power intensity emitted from the emitting antenna.

$\mathrm{P_{received}}$ is the power intensity received at the receiving antenna.

By performing these four different tests, the results provided a clear comparison between the shielding effectiveness of the uncoated and coated samples, allowing for the evaluation of which material and coating concentration offered the best performance.

4 Model creation and boundary conditions

In the COMSOL simulation model, a wire-mesh 3D geometry is created with a grid-type shielding structure, incorporating square grid elements of 100 mm × 150 mm and specified wire diameters. The mesh type tetrahedral or hexahedral is carefully selected to balance computational efficiency and accuracy. The shielding structure is positioned at variable distances from the central radiation source in each direction

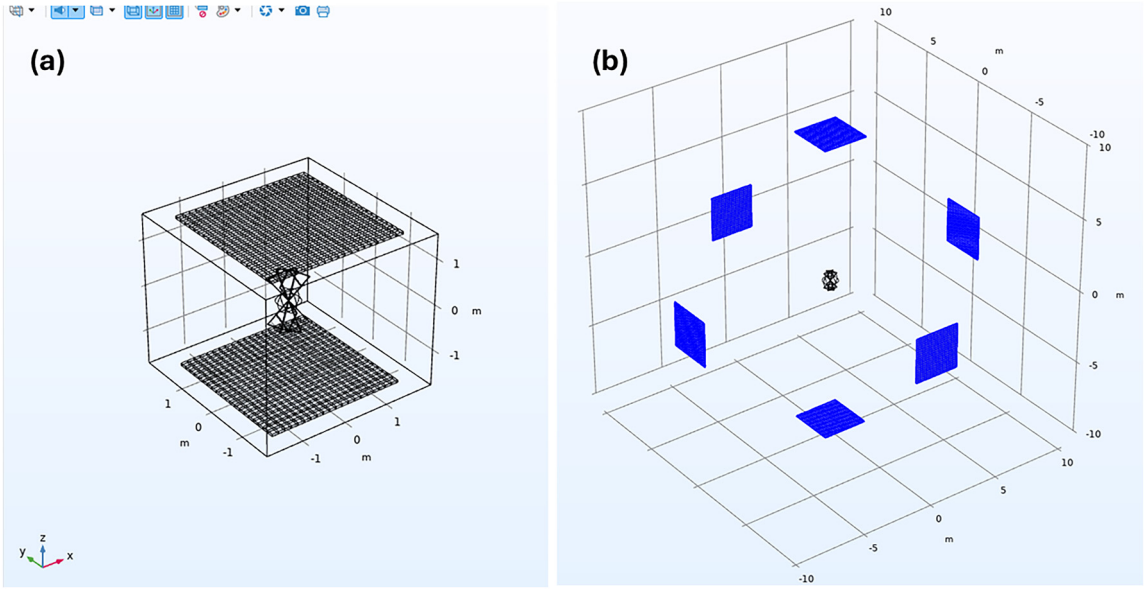


Figure 6: Grid type shielding design with variable distance between radiation source and shielding (A) grid design (B) isometric view of design for all four directions.

to analyze spatial effects on EMI attenuation. Material properties, including electrical conductivity, permittivity, and permeability, are assigned using the COMSOL Material Library, with additional user-defined materials for cases not available in the database. The model employs the frequency-domain form of Maxwell's equations to simulate electromagnetic wave propagation, as governed by the defined wave equations (Equations (1) and (2)). Perfect Electric Conductor (PEC) boundary conditions are applied across all surfaces and edges to ensure ideal reflection behavior. The electromagnetic radiation source is centrally located within the domain, and its properties such as type, frequency, and orientation are explicitly defined to simulate real-world EMI interaction with the shielding grid. To analyze the shielding performance under varying spatial conditions, a grid-type shielding design was developed with adjustable distances between the radiation source and the shielding surface. As illustrated in Figure 6A, the grid design allows systematic evaluation of shielding effectiveness based on positional variation. Figure 6B provides an isometric view of the design layout, showing the arrangement from all four directions to simulate diverse electromagnetic exposure scenarios.

The governing equation can be written in the form

$$\nabla \times (\mu_r^{-1} \nabla \times \boldsymbol{E}) - k_o^2 \varepsilon_{rc} \boldsymbol{E} = 0 \tag{1}$$

where $\boldsymbol{n}$ = refractive index, $\boldsymbol{E}$ = electric field intensity, ε_r = Relative permittivity

The Wave Number (k_o) of free space

$$k_o = \omega \sqrt{\varepsilon_0 \mu_0} = \frac{\omega}{c_0} \tag{2}$$

where c_0 is the speed of light in vacuum.

5 Results and discussions

5.1 Simulated results

The shielding effectiveness (SE) of the samples was numerically analyzed through simulation studies conducted using COMSOL Multiphysics. A model replicating the experimental setup was developed to evaluate the electromagnetic shielding performance of nickel foam samples coated with multi-walled carbon nanotubes (MWCNTs). The Electromagnetic Waves, Frequency Domain interface was used to simulate the response of the materials over a frequency range of 440–3,700 MHz. The simulation provided electric and magnetic field distributions, enabling precise computation of SE for each configuration.

In the simulation, the F1 sample coated with 20 mg/mL CNTs exhibited the highest shielding effectiveness, confirming the role of higher CNT concentration in enhancing electromagnetic attenuation. The SE for the F1 sample started at approximately 51.5 dB at 440 MHz and increased steadily to ~56.5 dB at 3,700 MHz. This performance is

attributed to improved electrical conductivity and surface area provided by the denser CNT network, promoting enhanced multiple reflections and absorption of incident radiation within the porous nickel matrix.

The F2 sample (16 mg/mL CNT) showed intermediate SE performance, ranging from ~50 dB to ~55 dB, while the uncoated nickel foam demonstrated a modest SE improvement from ~49 dB to ~54.5 dB, showcasing the intrinsic EMI shielding properties of the metallic matrix.

Interestingly, the simulation results showed a continuous increase in SE with frequency, contrary to the experimental observation, where SE peaked around 2 GHz and subsequently declined. This discrepancy can be attributed to several factors:

- The simulations employed Perfect Electric Conductor (PEC) boundaries, which assume infinite conductivity. While this simplifies computation, it overlooks the finite conductivity and lossy behavior of real-world composite materials, especially at higher frequencies, leading to an overestimation of SE.
- In the physical setup, edge diffraction and fringing fields around the sample introduce additional transmission pathways for EM waves, reducing the actual shielding performance. These edge-related losses are not inherently captured in the simulation unless explicitly modeled, leading to deviation in SE trends.
- At higher frequencies (>2 GHz), impedance mismatch between the test material and free space increases, resulting in partial reflection and less effective absorption. Additionally, the skin depth of the nickel foam becomes comparable to or smaller than the pore wall thickness, reducing its bulk attenuation capabilities – effects that are not fully accounted for in the simulation.

5.2 Experimental evaluation of shielding effectiveness (SE)

The shielding effectiveness (SE) of the developed samples was experimentally measured using a Vector Network Analyzer (VNA) inside an anechoic chamber. The VNA transmitted a controlled electromagnetic signal through an emitting antenna toward the test samples, while a receiving antenna recorded the transmitted signal after interaction with the material. The SE was calculated based on the difference between the transmitted and received signal strengths.

The SE values exhibited a clear frequency-dependent trend across the 800 MHz–3 GHz range. Initially, at lower frequencies, all tested samples including the uncoated nickel foam, F2 sample (coated with 16 mg/mL CNTs), and F1 sample (coated with 20 mg/mL CNTs) demonstrated relatively low SE values. As frequency increased, the shielding performance improved steadily for all configurations, with SE values peaking near 2 GHz. However, beyond this frequency, a decline in SE was observed across all samples.

This behavior can be attributed to the interaction between electromagnetic wave energy and material absorption characteristics. At lower frequencies, the energy of the incident radiation is insufficient for substantial interaction, leading to limited attenuation and lower SE. With increasing frequency, the energy of the radiation becomes sufficient to induce stronger interactions, resulting in increased absorption and reflection, and hence higher SE values. However, beyond 2 GHz, the limited physical dimensions of the sample (100 mm × 150 mm) may have led to edge effects or leakage, reducing the shielding capability. This suggests that increasing the sample size could potentially improve performance by providing a larger area for wave attenuation.

Among all tested configurations, the F1 sample (20 mg/mL CNT coating) demonstrated the highest shielding effectiveness, reaching a peak value of 48.165 dB. This superior performance is attributed to the higher concentration of multi-walled carbon nanotubes (MWCNTs), which enhanced electrical conductivity and facilitated effective electromagnetic wave attenuation through absorption and multiple reflections.

The F2 sample (16 mg/mL CNT coating) also showed improved SE compared to the uncoated sample, albeit slightly lower than F1, confirming the beneficial role of CNT integration (Figure 7). The uncoated nickel foam, while providing some inherent shielding due to its metallic porous

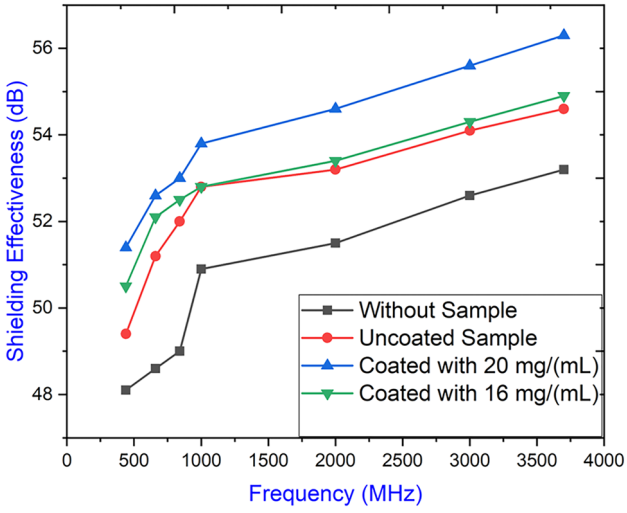


Figure 7: Effects of frequencies on the shielding effectiveness for all four samples.

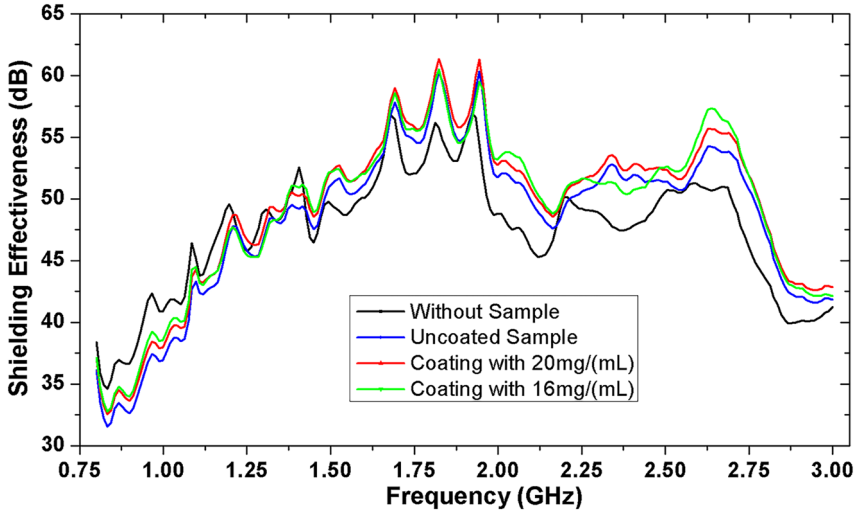


Figure 8: Shielding effectiveness verses frequency in GHz for four samples.

structure, was the least effective among the three, emphasizing the necessity of CNT coatings for enhanced performance. The variation of shielding effectiveness (SE) with frequency for the four tested samples is presented in Figure 8. The graph illustrates that SE generally increases with frequency, particularly between 0.8 GHz and 2.0 GHz, with the F1 sample (20 mg/mL CNT) achieving the highest SE throughout the range. This trend confirms the enhanced attenuation capabilities of CNT-coated nickel foam, especially at higher frequencies relevant to mobile communication and electronic device interference.

5.3 Discussion and literature correlation

The experimental and simulation results from this study demonstrate a strong correlation between carbon nanotube (CNT) concentration and enhanced shielding effectiveness (SE) in CNT-coated Cu-Ni foam composites. Both approaches consistently showed that increasing the CNT concentration improved the material's ability to attenuate electromagnetic radiation across a broad frequency range. These observations are well supported by prior research in the field.

Faisal et al. [19] reported that nickel foam, due to its intrinsic electrical conductivity, is a suitable substrate for EMI shielding applications. The open-cell porous structure facilitates both reflection and absorption of electromagnetic waves. This is consistent with the findings of Antunes et al. [20], who emphasized the role of conductive porous materials combined with carbon-based fillers to enhance multi-mechanism EMI shielding, including reflection, absorption, and scattering.

In this study, the F1 sample (20 mg/mL CNT concentration) exhibited the highest SE in both experimental (48.165 dB) and simulation (56.5 dB) results, significantly outperforming the F2 sample (16 mg/mL) and the uncoated Ni foam. This trend aligns closely with Varela et al. [21], who highlighted that higher CNT loadings lead to greater conductivity and more effective multi-path scattering, thereby improving EMI shielding performance.

The frequency-dependent behavior observed where SE increased with frequency up to ~2 GHz before tapering was clearly visible in the experimental results, attributed to energy absorption efficiency peaking at higher frequencies. The COMSOL simulation, however, predicted a continuous increase in SE up to 3,700 MHz. This difference may stem from idealized boundary conditions in simulations versus real-world signal leakage and geometric limitations in physical testing. Nevertheless, both findings corroborate the work of He et al. [22], who found that CNT incorporation enhances both reflection and absorption, particularly at higher frequencies.

The dip-coating technique used in this study proved efficient for uniform CNT deposition [23]. This supports findings by Ji et al. [24], who demonstrated that dip-coating offers good control over CNT dispersion. However, Zhang also cautioned about pore clogging at higher CNT concentrations, a concern addressed in this work by limiting the CNT content to maintain material breathability an important factor for applications such as window or door shielding panels.

Further validation of simulation reliability was observed through comparison with STOICA et al. [25], who employed electromagnetic solvers like Ansys HFSS to predict EMI shielding behavior. Their work supports the conclusion that simulation tools like COMSOL can accurately predict trends and peak SE values, thereby guiding material optimization and reducing experimental effort.

In terms of performance comparison, the 100 PPI Cu-Ni-CNT foam (1.5 mm thick) used in this study achieved a maximum SE of 53.4 dB, marking a 14 dB improvement over the uncoated sample. This performance is consistent with previous studies reporting that the shielding effectiveness (SE) of CNT-based conductive composites is strongly governed by CNT loading, electrical conductivity, foam porosity, and material thickness. Higher CNT concentrations promote the formation of conductive networks, while porous architectures enhance electromagnetic wave attenuation through multiple internal reflections and absorption mechanisms (Singh et al.; Lan et al.) [26, 27]. The open-cell metallic foam structure in the present work provided numerous internal reflection pathways, which, combined with the high electrical conductivity of MWCNTs, contributed to superior EMI shielding performance. Furthermore, carbon-fiber- and graphite-based composites have demonstrated moderate EMI shielding capability due to their conductive carbon framework; however, their conductivity and shielding efficiency generally remain lower than those of metal-based conductive foam systems (Bashir et al.) [28]. More advanced architectures, including SiC-coated carbon fibers and SiC nanofiber-coated carbon composites, have also exhibited promising EMI shielding behavior, primarily through enhanced dielectric loss and interfacial polarization mechanisms, although these systems often involve more complex fabrication routes and higher processing costs (Zhang et al.; Zhang and Cheng) [5, 29].

Ultimately, the combination of lightweight design, effective shielding, and cost-efficient fabrication makes the Cu-Ni-CNT composite a highly competitive material for EMI shielding applications. This study confirms that CNT-enhanced metal foams not only match but often exceed the SE of conventional materials, particularly when optimized for structure, thickness, and CNT loading.

6 Conclusions

This study successfully demonstrated the development and performance evaluation of a lightweight, multi-layered electromagnetic interference (EMI) shielding material using open-cell nickel (Ni) foam integrated with multi-walled carbon nanotubes (MWCNTs). Experimental tests, supported by COMSOL Multiphysics simulations, confirmed that the incorporation of CNTs enhances shielding effectiveness (SE) due to improved electrical conductivity and multi-path attenuation mechanisms. The shielding performance was found to be frequency-dependent and strongly influenced by CNT concentration.

Key conclusions:

- A lightweight and scalable EMI shielding composite was successfully developed using 100 PPI open-cell nickel foam coated with multi-walled carbon nanotubes (MWCNTs) via a simple dip-coating process, offering a cost-effective alternative to complex fabrication routes reported in prior studies.
- The incorporation of MWCNTs significantly enhanced the shielding effectiveness (SE) of the nickel foam due to the formation of an interconnected conductive network. Increased CNT loading resulted in improved EMI attenuation, with the F1 sample (20 mg/mL) consistently outperforming the F2 sample (16 mg/mL) and the uncoated foam.
- Experimentally, the F1 sample achieved the highest SE of 48.165 dB, representing an improvement of 0.545 dB over the uncoated foam and 0.235 dB over the F2 sample, while the F2 sample achieved an SE of 47.93 dB, corresponding to a 0.31 dB enhancement over the uncoated nickel foam. Although the difference between F1 and F2 is modest, it confirms the positive influence of CNT concentration on EMI shielding performance.
- All samples exhibited a frequency-dependent SE behavior within the 800 MHz–3 GHz range, with SE increasing up to approximately 2 GHz and then gradually declining. This trend was attributed to the interplay between electromagnetic wavelength, porous structure dimensions, impedance mismatch, and edge-related leakage effects at higher frequencies.
- COMSOL Multiphysics simulations successfully captured the relative performance trends among the samples, predicting a continuous increase in SE with frequency, with the F1 sample reaching a simulated SE of ~56.5 dB at 3,700 MHz. The deviation from experimental trends at higher frequencies was attributed to idealized PEC boundary conditions, neglected edge effects, and finite material conductivity, highlighting both the strengths and limitations of the numerical model.
- The EMI shielding mechanism of the Ni-CNT composite arises from a synergistic combination of reflection, absorption, and multiple internal scattering, facilitated by the high electrical conductivity of MWCNTs and the large surface area and interconnected porosity of the nickel foam.
- Owing to its lightweight nature, fabrication simplicity, ventilation capability, and broadband shielding performance, the developed Ni-CNT composite shows strong potential for practical EMI shielding applications, including architectural shielding for windows and doors, electronics packaging, smart buildings, IoT enclosures, and consumer electronic housings.

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